

INDIVIDUAL CASE SAFETY REPORT (ICSR)

PROCESSING: CASE 2

Section 1: Literature Intake Sheet

Article ID	IJMS241310919
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Author	Elena Giovannini, Maria Paola Bonasoni, Michele D'Aleo, Ione Tamagnini, Matteo Tudini, Paolo Fais, Susi Pelotti
Country	Italy
Patient Age	65 years old
Sex	Male
Drug	Pembrolizumab
Reaction	Myasthenia gravis, myocarditis, necrotizing myositis, acute kidney failure, hepatic centrolobular necrosis, death
Serious?	Yes
Reporter Type	Healthcare Professional / Pathologist (Post-mortem evaluation requested due to legal medical liability evaluation)
Source	Literature Case Report

Section 2: Case Triage

Patient?	Yes (65-year-old male)
Reporter?	Yes (Medical examiners / Pathologists from the University of Bologna and Azienda USL-IRCCS di Reggio Emilia)
Drug?	Yes (Pembrolizumab)
Drug reaction?	Yes (Myasthenia gravis, myocarditis, myositis, death)

Triage Decision: VALID ICSR. All four minimum criteria for an Individual Case Safety Report are fulfilled.

Section 3: Data Extraction Sheet

Patient	CASE002
Age	65 years
Gender	Male
Weight	Not reported
History	Nodular melanoma of the left forearm (pT4a, Breslow thickness 12 mm, mitoses ≥ 10 mm ² , vertical growth phase, positive deep surgical margin; NRAS mutations detected: Q61K, Q61R, Q61L, Q61H); Left axillary lymphadenectomy confirmed 1/4 positive lymph node (pN1b).
Medical history	Coronary artery disease with severe diffuse wall calcification and atherosclerosis (80% luminal occlusion of the left main trunk).
Suspect drug	Pembrolizumab
Dose	200 mg

Frequency	Every 3 weeks (Single cycle administered)
Route	Intravenous
Lot number	Not reported
Start date	Approximately 3 months after histological diagnosis
Stop date	Discontinued after 1 dose due to symptom onset
Concomitant drugs	Not reported for the oncology regimen; therapy for adverse events included methylprednisolone (60 mg IV QD), pyridostigmine (60 mg PO TID), and anticoagulants.
Lab values	Serum creatinine: 2.49 mg/dL, GFR: 23 mL/min/1.73 m ² , ALT: 659 U/L, AST: 1687 U/L, CRP: 7.9 mg/dL, Troponin I: 25,010 ng/L, Creatine Kinase (CK): 14,977 U/L.
Reaction	Pembrolizumab-induced triad of Myasthenia Gravis, Lympho-histiocytic Myocarditis, and Necrotizing Myositis, complicated by acute kidney injury, hepatic injury, respiratory failure, atrioventricular block, and fatal ventricular tachycardia.
Outcome	Fatal (Death occurred 15 days after the single dose).
Action taken	Drug permanently withdrawn.
Dechallenge	Negative (Condition rapidly deteriorated despite drug withdrawal and medical management).
Rechallenge	Not performed.
Reporter assessment	The investigators concluded that the acute onset of overlapping myasthenia gravis, myocarditis, and necrotizing myositis represents a rare, predictable but non-preventable immune-related adverse event (irAE) directly caused by pembrolizumab therapy.

Section 4: Narrative Writing

Clinical Narrative

Case Summary:

A 65-year-old male patient with advanced metastatic nodular melanoma (pT4a pN1b, NRAS-mutated) experienced a fatal overlapping systemic immune-related adverse event (irAE) triad consisting of myasthenia gravis, necrotizing myositis, and severe lympho-histiocytic myocarditis following the administration of a single adjuvant cycle of pembrolizumab (200 mg).

Onset and Clinical Progression:

Twelve days after receiving the first dose of pembrolizumab, the patient developed a fever (37.8 deg C), severe muscle aches, and constipation. The clinical state rapidly worsened with the development of diplopia and gait instability. On day 15 post-infusion, he was hospitalized. Neurological assessment confirmed impaired lateral rotation of the right eye, left eyelid ptosis, and reduced motor strength in the lower limbs, raising immediate suspicion of myasthenia gravis and acute myositis.

Management and Diagnostic Monitoring:

The patient was immediately treated with intravenous methylprednisolone (60 mg daily) and oral pyridostigmine (60 mg three times daily). Initial admission laboratories and ECG demonstrated:

- **Cardiovascular:** Sinus rhythm accompanied by a right bundle branch block. Echocardiogram revealed a preserved left ventricular ejection fraction (>60%).
- **Renal/Hepatic:** Acute kidney injury with serum creatinine at 2.49 mg/dL (GFR: 23 mL/min/1.73 m²), profound transaminitis (AST: 1687 U/L, ALT: 659 U/L), and systemic inflammation (CRP: 7.9 mg/dL).

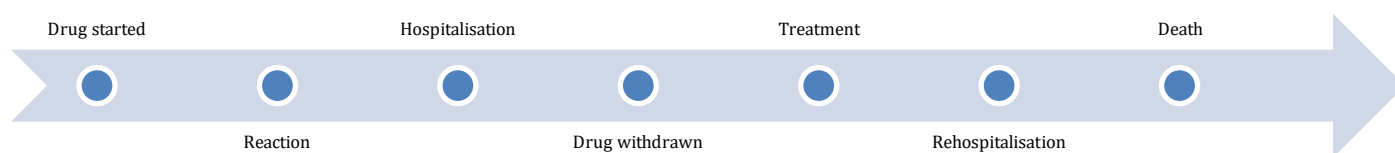
On the second hospitalization day, the patient experienced acute palpitations and severe dyspnea, showing an onset of atrial fibrillation. Laboratory monitoring demonstrated extreme elevations of cardiac markers: Troponin I rose to 25,010 ng/L and Creatine Kinase spiked to 14,977 U/L, solidifying a diagnosis of concurrent fulminant myocarditis. Anticoagulant therapy was added.

On the third day of hospitalization, cardiac monitoring revealed progress to a second-degree Mobitz Type 2 atrioventricular block. The clinical course rapidly deteriorated into combined respiratory and renal failure, prompting non-invasive mechanical ventilation (FiO₂ 40%) and continuous renal replacement therapy (dialysis). Despite maximum supportive measures, the patient developed refractory hypoxemia and sustained ventricular tachycardia with peak torsades de pointes fibrillation. Resuscitative maneuvers were unsuccessful, and the patient died 15 days after receiving the pembrolizumab dose.

Autopsy and Histopathological Findings:

Post-mortem examination demonstrated heavy lungs with marked congestion and bilateral serosanguinous pleural effusions. The heart was globally enlarged (530 g) with a marbled, dyschromic left ventricular myocardium and co-existing severe coronary artery atherosclerosis (80% occlusion of the left main trunk). Microscopic analysis confirmed:

- **Myocardium:** Diffuse, severe lympho-histiocytic myocarditis with contraction band necrosis. Immunohistochemistry showed dense CD3+ and CD8+ cytotoxic T-cell infiltrates (Tia-1+, PD-1+) outnumbering CD4+ helper cells, abundant CD68/PGM1+ macrophages, and robust focal upregulation of PD-L1 on damaged cardiomyocyte membranes.
- **Skeletal Muscle (Iliopsoas):** Severe necrotizing myositis with prominent lymphocytic infiltration (CD3+, equal CD4+ to CD8+ ratio), high quantities of CD20+ B-cells, and focal PD-L1 expression on injured myofibers and surrounding macrophages.
- **Other Organs:** Diffuse acute tubular necrosis in the kidneys, severe centrilobular hepatic necrosis, and focal acute bronchopneumonia in the right lung.



Section 5: MedDRA Coding

Lowest Level Term (LLT)	Myasthenia gravis / Immune-mediated myocarditis / Immune-
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	mediated myositis
Preferred Term (PT)	Myasthenia gravis / Myocarditis / Myositis
High Level Term (HLT)	Myasthenia gravis / Noninfectious myocarditis / Myopathies
High Level Group Term (HLGT)	Neuromuscular junctions and muscle nerve disorders / Myocardial disorders / Muscle disorders
System Organ Class (SOC)	Nervous system disorders / Cardiac disorders / Musculoskeletal and connective tissue disorders

****MedDRA numeric codes omitted because the terminology is licensed; hierarchy shown for educational purposes**

Section 6: Seriousness

Seriousness: Yes

Reason for Seriousness:

☒	Death (Patient died 15 days post-treatment)
☒	Life-threatening (Fulminant ventricular tachycardia, respiratory failure)
☒	Hospitalisation (Admitted for acute neurological and cardiac monitoring)
☐	Congenital anomaly
☐	Disability
☒	Other medically important condition (Fulminant overlapping irAE triad)

Section 7: Expectedness

Classification: Expected

Evidence:

According to reference safety information (USPI and SmPC) for pembrolizumab :

- **Immune-Mediated Myocarditis:** Explicitly detailed under Section 5 (Warnings and Precautions) of the USPI and Section 4.4 (Special Warnings) of the SmPC. Product labeling explicitly warns that immune checkpoint inhibitors can cause severe, fatal immune-mediated myocarditis (incidence <1%). Immediate permanent discontinuation and high-dose corticosteroid management are advised.
- **Immune-Mediated Myositis and Myasthenia Gravis:** Evaluated under neuromuscular/musculoskeletal immune-related adverse events. Labeling notes post-marketing reports of systemic myositis, rhabdomyolysis, and myastheniform syndromes under Section 6 (Adverse Reactions) and Section 4.8 of the SmPC. The overlapping presentation of these distinct target organ toxicities is documented within the clinical safety profile.

Section 8: Causality Assessment

WHO-UMC Causality Category: Probable / Likely

Reason: The toxicities developed within a highly characteristic, plausible biological timeline (12 days after a single cycle), presented as a classic documented overlapping immune-mediated triad unique to checkpoint inhibitor mechanisms, and cannot be reasonably explained by the progression of the underlying metastatic melanoma.

Naranjo Score Assessment

1. Previous conclusive reports on this reaction?	Yes (+1) — Pembrolizumab is well-established to induce myocarditis and myositis.
2. Did the adverse event occur after the suspected drug was administered?	Yes (+2) — Symptoms occurred 12 days following the initial infusion.
3. Did the adverse reaction improve when the drug was discontinued?	No (0) — The patient continuously deteriorated and died despite drug withdrawal.
4. Did the adverse reaction reappear when the drug was readministered?	No / Not Done (0) — Rechallenge was not attempted due to fatality.
5. Are there alternative causes that could have on their own caused the reaction?	No (+2) — Metastatic melanoma does not spontaneously produce simultaneous autoimmune myocarditis, myositis, and myasthenia gravis.
6. Did the reaction reappear when a placebo was given?	No / Not Done (0)
7. Was the drug detected in the blood in concentrations known to be toxic?	No / Not Done (0)

8. Was the reaction more severe when the dose was increased / less severe when decreased?	No / Not Done (0)
9. Did the patient have a similar reaction to the same/similar drugs in previous exposure?	No / Unknown (0)
10. Was the adverse event confirmed by any objective evidence?	Yes (+1) — Confirmed via highly elevated biomarkers (Troponin I: 25,010 ng/L, CK: 14,977 U/L) and direct post-mortem histopathology and immunohistochemistry.

Total Naranjo Score: +6

Causality Interpretation: Probable (Score of 5–8 indicates a probable causal relationship).

****For immune checkpoint inhibitor toxicities, the Naranjo algorithm has limitations.**

Section 9: Quality Check

Missing patient info	None (Age, sex, histology, and clinical trajectory fully detailed)
Dates logical	Yes (Onset 12 days post-dose, admission day 15, death day 15)
Duplicate events	Checked; no duplicate records found
Narrative complete	Pass (Reviewer initials: MJ, 11 July, 2026)
MedDRA verified	Yes
Seriousness correct	Yes
Causality complete	Yes
CIOMS Form	Complete